

Official Statistics

A Textbook for FYUGP Programme
Vth Sem BSc Statistics,
Vimala College (Autonomous), Thrissur

Dr Jerin Paul
Assistant Professor
Department of Statistics
Vimala College (Autonomous), Thrissur

June 2026

Contents

Preface	5
Acknowledgements	5
Course Information	5
Course Details	5
Credit Structure	6
Detailed Syllabus	6
Books and References	7
1 Introduction	9
1.1 Introduction to Indian and International Statistical systems.	9
1.1.1 The Indian Statistical System: An In-Depth Overview	9
1.1.2 The International Statistical System: An In-Depth Overview	12
1.2 Methods of Collection of Official Statistics	13
1.2.1 Full Field Enumeration (The Traditional Census)	14
1.2.2 Large-Scale Sample Surveys	14
1.2.3 The Register-Based Approach (Administrative Data)	15
1.2.4 Combined Census Methodologies	15
1.2.5 The Rolling Census Approach	16
1.2.6 Modern Data Capture Techniques	16
1.3 Role, function and activities of Central and State Statistical organizations. .	17
1.4 Organization of large-scale sample surveys. Role of Ministry of Statistics & Programme Implementation (MoSPI).	17
1.5 Central Statistical Office (CSO), National Sample Survey Office (NSSO), and National Statistical Commission.	17
1.6 Government of India's principal publications containing data on topics such as population, industry and finance.	17
1.7 Scope and contents of population census of India.	17
2 Population and Economic Statistics	19
2.1 Population growth in developed and developing countries.	19
2.2 Evaluation of performance of family welfare programmes.	19
2.3 Projections of labour force and manpower.	19
2.4 Statistics related to industries, foreign trade, balance of payment.	19
2.5 Statistics related to cost of living, inflation, educational and other social statistics.	19

3	Economic Development and National Income	21
3.1	Economic development.	21
3.2	Growth in per capita income and distributive justice; indices of development.	21
3.3	Human Development Index.	21
3.4	National income estimation — Product approach.	21
3.5	National income estimation — Income approach.	21
3.6	National income estimation — Expenditure approach.	21
4	Measuring Inequality and Poverty	23
4.1	Measuring inequality in incomes: Lorenz curve.	23
4.2	Gini Coefficient.	23
4.3	Theil's measure.	23
4.4	Poverty measurements: Different issues.	23
4.5	Measures of incidence and intensity of poverty.	23
	Glossary	25
	References	27

Preface

This textbook has been written to meet the requirements of the **Four Year Undergraduate Programme (FYUGP)** of Calicut University for the course on **Official Statistics**.

The book is organised into four chapters aligned with the Calicut University FYUGP syllabus:

Chapter	Title
1	Introduction
2	Population and Economic Statistics
3	Economic Development and National Income
4	Measuring Inequality and Poverty

Each chapter opens with learning objectives. Key terms are **bolded** when first introduced.

Acknowledgements

The author acknowledges data sources used in this book: the Ministry of Statistics and Programme Implementation (MoSPI), the Registrar General of India, the Reserve Bank of India, and the United Nations Statistics Division.

Course Information

Course Details

Programme	B.Sc. Statistics
Course Code	STA5EJ305
Course Title	Official Statistics
Type of Course	Major Elective
Semester	V
Academic Level	300–399

Credit Structure

Credit	Lecture/Week	Tutorial/Week	Practical/Week	Total Hours
4	4	—	—	60

Pre-requisites: Basic Statistics

Course Objective: Aware students of the role of Statistics in National Policy Formulation, Planning, and framing of various policies by the Governments.

Detailed Syllabus

Module I — Introduction (12 Hrs, 20 Marks)

1. Introduction to Indian and International Statistical systems.
2. Methods of collection of official statistics.
3. Role, function and activities of Central and State Statistical organizations.
4. Organization of large-scale sample surveys. Role of Ministry of Statistics & Programme Implementation (MoSPI).
5. Central Statistical Office (CSO), National Sample Survey Office (NSSO), and National Statistical Commission.
6. Government of India's principal publications containing data on topics such as population, industry and finance.
7. Scope and contents of population census of India.

Module II — Population and Economic Statistics (12 Hrs, 20 Marks)

8. Population growth in developed and developing countries.
9. Evaluation of performance of family welfare programmes.
10. Projections of labour force and manpower.
11. Statistics related to industries, foreign trade, balance of payment.
12. Statistics related to cost of living, inflation, educational and other social statistics.

Module III — Economic Development and National Income (12 Hrs, 15 Marks)

13. Economic development.
14. Growth in per capita income and distributive justice; indices of development.
15. Human Development Index.
16. National income estimation — Product approach.
17. National income estimation — Income approach.
18. National income estimation — Expenditure approach.

Module IV — Measuring Inequality and Poverty (12 Hrs, 15 Marks)

19. Measuring inequality in incomes: Lorenz curve.
20. Gini Coefficient.
21. Theil's measure.
22. Poverty measurements: Different issues.
23. Measures of incidence and intensity of poverty.

Books and References

1. Guide to Official Statistics, CSO (1999).
2. Statistical System in India, CSO (1995).
3. Principles and Accommodation of National Population Census, UNESCO.
4. Monthly Statistics of Foreign Trade in India, DGCIS, Calcutta and other Government Publications.
5. Keyfitz, N. (1977). *Applied Mathematical Demography*. Springer Verlag.
6. Sen, A. (1977). *Poverty and Inequality*.

7. Chubey, P.K. (1995). *Poverty Measurement*. New Age International.

Chapter 1

Introduction

1.1 Introduction to Indian and International Statistical systems.

1.1.1 The Indian Statistical System: An In-Depth Overview

The Indian Statistical System is a vast, complex, and highly structured network responsible for capturing the socio-economic and demographic realities of a nation with over 1.4 billion people. It operates within the broader administrative and federal framework of the country.

1.1.1.1 Historical Evolution

- **Ancient and Medieval Roots:** Statistics in India are as old as statecraft itself. The ancient Indian treatise, *Arthashastra*, explicitly refers to a network of village-level accountants responsible for collecting data on economic output. Similarly, the medieval text *Ain-i-Akbari* details an exhaustive administrative apparatus used to collect data on farm produce and standardizing weights and measurements.
- **The British Raj (1858–1947):** The modern official statistical system began taking shape during colonial rule. However, colonial efforts were primarily driven by self-interest—specifically to track markets for English products and manage revenue. Consequently, trade statistics were highly developed, while data on domestic socio-

economic development was neglected. British officials also utilized these administrative statistics as propaganda to demonstrate to the British Parliament how well they were governing India.

- **Post-Independence and the Mahalanobis Era:** Following independence in 1947, India recognized the pressing need for a robust database to drive economic planning. A serious effort to revamp the system was led by **Prof. P.C. Mahalanobis**, who introduced the widespread use of random sampling—a revolutionary concept at the time. Under his guidance, the Directorate of National Sample Survey was established in 1950, followed by the Central Statistical Organization (CSO) in 1951, laying the modern foundation of India’s statistical infrastructure.
- **Modern Reforms:** In 2000, the government appointed a commission led by C. Rangarajan to identify system deficiencies. This directly led to the establishment of the **National Statistical Commission (NSC)** in 2006 as the apex advisory body on statistical matters, and the enactment of the comprehensive **Collection of Statistics Act, 2008**.

1.1.1.2 Constitutional and Legal Framework

The Indian Constitution dictates how statistical responsibilities are shared. In the Seventh Schedule, the subject of ‘Statistics’ is covered in both the **Union List (Entry 94)** and the **Concurrent List (Entry 45)**. Because it is on the Concurrent List, both the Central and State Governments have the authority to operate and collect data in this domain.

Critical legislative backing for the system includes:

- **The Census Act, 1948:** Empowers the collection of demographic data.
- **The Registration of Births and Deaths Act, 1969:** Mandates the Civil Registration System.
- **The Collection of Statistics Act, 2008:** Arms statistical officers with the power to collect a wide array of socio-economic and industrial data from enterprises and citizens.

1.1.1.3 Structural Organization: Decentralization

The most defining feature of the Indian Statistical System is that it is **laterally and vertically decentralized**.

- **Lateral Decentralization (Central Level):** At the national level, data collection is divided among various line Ministries and Departments based on their administrative subjects (e.g., the Ministry of Agriculture collects crop data; the Ministry of Commerce collects trade data).
- **Vertical Decentralization (Centre and States):** Authority is shared between the Union Government and the States/Union Territories. State statistical organizations collect and generate data for sectors under their purview, and the Centre acts as the coordinating agency to present this data on an all-India basis.

1.1.1.4 Key Institutions at the Centre and State

1. Ministry of Statistics and Programme Implementation (MoSPI): MoSPI is the nodal agency for the planned development of the statistical system. It oversees the **National Statistical Office (NSO)**, which was formed in May 2019 by merging the Central Statistics Office (CSO) and the National Sample Survey Office (NSSO) to streamline operations.

- **Central Statistics Office (CSO):** Responsible for macroeconomic data. Its major activities include compiling National Income Accounting (GDP), processing the Annual Survey of Industries (ASI), conducting Economic Censuses, compiling the Index of Industrial Production (IIP), and setting statistical standards.
- **National Sample Survey Office (NSSO):** India's premier agency for conducting large-scale, nationwide sample surveys. It collects primary data on socio-economic subjects such as employment, consumer expenditure, health, and agriculture.

2. State Directorates of Economics and Statistics (DES): At the apex of the state level is the DES. They coordinate all statistical activities within the state, maintaining uniformity with Central standards. They compute the State Domestic Product (SDP), compile local price indices, and participate in national surveys on a matching-sample basis. The DES operates through District Statistical Offices, extending all the way down to block and village levels.

1.1.2 The International Statistical System: An In-Depth Overview

Because statistics are essential for sustainable economic, environmental, and social development, the international community maintains a highly coordinated system to ensure data comparability, share methodologies, and provide technical assistance globally.

1.1.2.1 The United Nations Statistical Division (UNSD)

The UNSD acts as the central hub of the global statistical system. Its primary responsibilities are: 1. **Global Coordination:** Directing global statistical endeavors and ensuring that definitions and concepts are standardized so that data from India can be accurately compared to data from Brazil, the US, or Japan. 2. **Data Repository:** Serving as the central global repository for gathering, analyzing, and publishing statistics across every field for domestic and international use.

1.1.2.2 UN Regional Commissions

To facilitate cooperation among countries in specific geographic regions, the UN operates Regional Commissions. They implement global standards locally and offer technical support to participating governments.

- India is a prominent member of the **Economic and Social Commission for Asia and the Pacific (ESCAP)**.
- Other commissions include the ECE (Europe), ECA (Africa), ECLAC (Latin America/Caribbean), and ESCWA (Western Asia).

1.1.2.3 Specialized International Agencies

The international system delegates specific statistical domains to specialized, independent agencies. These agencies set the ultimate global standards for their respective fields:

- **International Labour Organization (ILO):** Sets standards for labour, employment, and wage statistics.

- **Food and Agriculture Organization (FAO):** Manages global statistics regarding agriculture, forestry, and fisheries.
- **International Monetary Fund (IMF) & World Bank:** Handle international financial statistics, balance of payments, and global economic development data.
- **World Health Organization (WHO):** Standardizes health and mortality statistics (such as the International Classification of Diseases).
- **UNESCO:** Manages global education and cultural statistics.

1.1.2.4 The UN Fundamental Principles of Official Statistics (UNFPOS)

To ensure that governments around the world produce statistics that citizens and the international community can trust, the UN established the UNFPOS. India formally adopted these principles in 2016. The core tenets include:

- **Relevance, Impartiality, and Equal Access:** Official statistics are a public good and must be disseminated objectively and impartially.
- **Professionalism and Scientific Principles:** Methods for data collection and processing must be strictly based on scientific principles and professional ethics, preventing political manipulation.
- **Confidentiality:** Individual data collected for statistical purposes (whether from a person or a business) must be kept strictly confidential and used exclusively for statistical analysis.
- **Transparency:** Statistical agencies must make their sources, methodologies, and procedures transparent to the public.

1.2 Methods of Collection of Official Statistics

The collection of official statistics involves structured methodologies designed to gather accurate, comprehensive, and reliable data about a country's population, economy, and social conditions. Statistical systems generally rely on the following primary approaches:

1.2.1 Full Field Enumeration (The Traditional Census)

The traditional census approach comprises a comprehensive operation of actively collecting information from individuals and households on a range of topics at a specified time. It involves canvassing the entire country, reaching every single household, and collecting individual information within a brief, stipulated period.

- **Essential Features:** This method is characterized by **individual enumeration** (each person is recorded separately), **universality** (covers the precisely defined national territory completely), **simultaneity** (data refers to a specific, well-defined point in time, known as the census moment), and defined **periodicity** (usually conducted decennially).
- **Advantages:** It provides an unparalleled benchmark snapshot of the entire population at a specific period. It is uniquely capable of generating detailed data for the lowest geographic and administrative domains (small-area statistics), which is critical for local-level planning and the delimitation of electoral boundaries.
- **Limitations:** It is the most elaborate, complex, and costly data collection activity a national statistical office undertakes. Because of the massive financial and administrative resources required, it is typically conducted only once every ten years. Consequently, the data tends to become outdated during the intercensal years.

1.2.2 Large-Scale Sample Surveys

Because of the rapid changes in socio-economic characteristics and the high cost of full censuses, statistical systems rely on continuing programmes of intercensal household sample surveys.

- **Methodology:** Information is collected from a scientifically selected, representative subset of the population (the sample) rather than the entire population. The population census typically serves as the primary sampling frame for these surveys.
- **Advantages:** Sample surveys are highly cost-effective and allow for the collection of detailed data on complex topics that are not suitable for a full census, such as labour force participation, consumer expenditure, poverty, and detailed industrial activities.
- **Limitations:** Estimates derived from sample surveys are subject to sampling errors.

Furthermore, because they rely on a smaller fraction of the population, they rarely provide reliable estimates for very small geographic areas or rare sub-populations.

1.2.3 The Register-Based Approach (Administrative Data)

In recent decades, there has been a significant shift toward utilizing existing administrative sources to produce official statistics.

- **Methodology:** This approach extracts data from administrative registers that are primarily maintained for government operations, such as civil registration (births and deaths), tax records, employment registers, and education enrolment. To effectively compile census-like statistics, these individual registers are linked at the unit level, which typically requires the existence of a universal personal identification number for accurate record linkage.
- **Advantages:** It significantly reduces the long-term costs of data collection and completely eliminates the response burden on the public. It also allows for the generation of statistical data at a much higher frequency (e.g., annually) compared to traditional censuses.
- **Limitations:** The definitions, concepts, and reference periods in administrative registers are determined by administrative rules, which may not align with international statistical standards. Furthermore, individuals outside the formal administrative system (e.g., undocumented migrants or homeless populations) may be entirely excluded, leading to coverage errors.

1.2.4 Combined Census Methodologies

To optimize resources and data quality, many national statistical organizations use a combined approach, integrating administrative registers with field enumeration or sample surveys.

- **Methodology:** Data from existing administrative registers are used directly as a source for certain statistical variables. For variables missing in the registers, or to verify existing data, the statistical office conducts ad-hoc sample surveys or partial field enumerations. In some instances, the statistical office will use register data to

pre-fill census questionnaires, and the enumerator or respondent only needs to verify the information and answer any remaining questions.

- **Advantages:** This hybrid approach reduces the length of the questionnaire and the burden on respondents while ensuring high-quality, comprehensive data collection.

1.2.5 The Rolling Census Approach

A rolling census represents a departure from the traditional model by conducting a continuous, cumulative survey that covers the whole country over a prolonged period (generally years), rather than a single short period of enumeration.

- **Methodology:** A fraction of the population is sampled and surveyed each year. The data collected over multiple years (e.g., a five-year cycle) are then statistically cumulated and modelled to produce annual population estimates and small-area statistics.
- **Advantages:** This approach smooths out the massive financial and labour peaks required for a traditional census and provides more frequently updated data.
- **Limitations:** It requires highly complex sampling frames, continuous tracking of addresses, and sophisticated mathematical modelling techniques to estimate the data accurately across time periods.

1.2.6 Modern Data Capture Techniques

The collection of official statistics now utilizes multi-mode approaches and advanced technology to improve accuracy and efficiency.

- **Computer-Assisted Interviewing:** Traditional paper questionnaires are being replaced by Computer-Assisted Personal Interviewing (CAPI) on portable devices (tablets/smartphones), Computer-Assisted Telephone Interviewing (CATI), and Internet-based self-enumeration. These systems have built-in logic controls to automatically check for consistency, skip irrelevant questions, and prevent missing values during the interview.
- **Optical Reading Technology:** For paper-based data collection, statistical offices use Optical Mark Reading (OMR), Optical Character Reading (OCR), and Intelligent

1.3. ROLE, FUNCTION AND ACTIVITIES OF CENTRAL AND STATE STATISTICAL ORGANIZATIONS

Character Recognition (ICR) to digitally scan and convert handwritten responses directly into computer data, drastically reducing manual data entry errors.

- **Geospatial Technology:** Data collection is increasingly supported by Geographic Information Systems (GIS) and Global Navigation Satellite Systems (GNSS). Enumerators use these to capture the exact geographic coordinates of households to ensure complete coverage and prevent duplication.

1.3 Role, function and activities of Central and State Statistical organizations.

1.4 Organization of large-scale sample surveys. Role of Ministry of Statistics & Programme Implementation (MoSPI).

1.5 Central Statistical Office (CSO), National Sample Survey Office (NSSO), and National Statistical Commission.

1.6 Government of India's principal publications containing data on topics such as population, industry and finance.

1.7 Scope and contents of population census of India.

Chapter 2

Population and Economic Statistics

- 2.1 Population growth in developed and developing countries.
- 2.2 Evaluation of performance of family welfare programmes.
- 2.3 Projections of labour force and manpower.
- 2.4 Statistics related to industries, foreign trade, balance of payment.
- 2.5 Statistics related to cost of living, inflation, educational and other social statistics.

Chapter 3

Economic Development and National Income

3.1 Economic development.

3.2 Growth in per capita income and distributive justice; indices of development.

3.3 Human Development Index.

3.4 National income estimation — Product approach.

3.5 National income estimation — Income approach.

3.6 National income estimation — Expenditure approach.

Chapter 4

Measuring Inequality and Poverty

4.1 Measuring inequality in incomes: Lorenz curve.

4.2 Gini Coefficient.

4.3 Theil's measure.

4.4 Poverty measurements: Different issues.

4.5 Measures of incidence and intensity of poverty.

Glossary

Age-Specific Fertility Rate (ASFR): The number of live births per 1,000 women in a specific age group in a given year.

Balance of Payments (BoP): A systematic record of all economic transactions between residents of a country and the rest of the world during a given period.

Census: A complete enumeration of all persons in a country at a specified point in time.

Civil Registration System (CRS): The continuous and permanent recording of vital events (births, deaths, marriages) by government authorities.

Consumer Price Index (CPI): A measure of the average change over time in the prices paid by consumers for a representative basket of goods and services.

Crude Birth Rate (CBR): The number of live births per 1,000 mid-year population in a given year.

Crude Death Rate (CDR): The number of deaths per 1,000 mid-year population in a given year.

GDP (Gross Domestic Product): The total monetary value of all final goods and services produced within a country's territory in a given period.

GNP (Gross National Product): GDP plus net factor income from abroad.

Index of Industrial Production (IIP): A composite indicator measuring short-term changes in the volume of production in the mining, manufacturing, and electricity sectors.

Infant Mortality Rate (IMR): The number of deaths of children under one year of age

per 1,000 live births in a given year.

Life Table: A statistical model that traces a hypothetical cohort through successive ages, recording mortality experience and life expectancy.

Maternal Mortality Ratio (MMR): The number of maternal deaths per 100,000 live births in a given period.

National Statistical Office (NSO): The apex body for official statistics in India, formed in 2019 by merging the CSO and NSSO.

Official Statistics: Quantitative or qualitative information compiled and disseminated by government agencies for public use.

Sample Registration System (SRS): A large-scale demographic survey providing annual estimates of fertility and mortality rates in India.

System of National Accounts (SNA): The international standard framework for national income accounting, currently SNA 2008.

Total Fertility Rate (TFR): The average number of children a woman would have if she experienced the current age-specific fertility rates throughout her reproductive years.

Wholesale Price Index (WPI): An index measuring price changes at the whole-sale/producer level.

References